

Physics-Based Transfer Learning on Data-Scarce Air Pollution

1st Khaled Saifullah Karim

Dept. of Computer Science and Engineering
BRAC University
Dhaka, Bangladesh
khaled.saifullah.karim@g.bracu.ac.bd

2nd Fawzia Hasan Orodi

Dept. of Computer Science and Engineering
BRAC University
Dhaka, Bangladesh
fawzia.hasan.oroedi@g.bracu.ac.bd

3rd Ahnaf Awsaf

Dept. of Computer Science and Engineering
BRAC University
Dhaka, Bangladesh
ahnaf.awsaf@g.bracu.ac.bd

I. ABSTRACT

This paper presents a physics-based student-teacher transfer learning framework for PM_{2.5} [8] air quality prediction under data-scarce conditions. A teacher LSTM model is trained on 34,139 hourly records from Beijing, China, using three physics-informed features — PM_{2.5} concentration, ambient temperature, and dew point [10] temperature — derived from a common meteorological schema. Its learned representations are transferred to a student model deployed in Kihumo, Kenya, where only 1,058 hourly records (44 days) are available, representing a 95.9% reduction in data compared to conventional requirements. Knowledge transfer is achieved through selective layer freezing: the LSTM layer is frozen to preserve universal atmospheric physics while only the Dense output layer (65 parameters) is fine-tuned on local data. The proposed method achieves an RMSE of 7.98 $\mu\text{g}/\text{m}^3$ and R^2 of 0.685 on the Kihumo test set, representing a 47.1% improvement over a from-scratch LSTM baseline (RMSE: 15.10 $\mu\text{g}/\text{m}^3$, R^2 : -0.127). Among all evaluated models, Random Forest [2] achieved the lowest RMSE of 6.60 $\mu\text{g}/\text{m}^3$, highlighting the competitiveness of ensemble methods on small datasets. An extended Bidirectional LSTM (BiLSTM) transfer variant is also evaluated. The findings demonstrate the viability of physics-informed transfer learning for rapid, low-resource air quality monitoring in data-scarce developing regions.

Keywords: Air pollution prediction, transfer learning, physics-informed learning, student-teacher model, data scarcity, LSTM, BiLSTM [4], selective layer freezing, PM_{2.5}, air quality forecasting.

A. Introduction:

Air pollution poses severe risks to human health, ecosystems, and climate. Pollutants such as PM_{2.5} [8], PM₁₀, NO₂, and O₃ are linked to respiratory and cardiovascular diseases. Accurate spatiotemporal prediction of air quality enables early warning systems and informed policy-making. Traditional air quality forecasting relies on numerical chemical transport models, which are computationally expensive and require detailed emission inventories. In contrast, data-driven machine learning (ML) and deep learning (DL) approaches

have gained popularity due to their predictive power and efficiency.

Despite their success, ML and DL models typically require large volumes of labeled data, which are often unavailable in developing countries or newly deployed monitoring regions. This data scarcity severely degrades model performance and generalization. Transfer learning has emerged as a promising solution by leveraging knowledge from data-rich source domains to improve learning in data-scarce target domains.

This paper proposes a physics-based transfer learning framework for air pollution prediction. The core idea is to combine atmospheric physics knowledge with a student-teacher deep learning architecture. The teacher model is trained using data-rich Beijing records and physics-informed features, while the student model learns from limited local Kihumo data by inheriting the teacher's frozen LSTM representations and fine-tuning only the output layer. This approach improves prediction accuracy, stability, and interpretability.

B. The main contributions of this work are:

- 1) A physics-informed student-teacher transfer learning framework for air quality prediction, implemented via selective LSTM layer freezing and output Dense layer fine-tuning.
- 2) Integration of domain knowledge through physics-motivated feature construction - specifically, deriving dew point temperature from temperature and relative humidity - to produce a common three-feature schema (PM_{2.5}, TEMP, DEWP) across both source and target domains.
- 3) A feature space alignment strategy that reuses the teacher's Min-Max scaler [1] on student data without re-fitting, ensuring the frozen LSTM receives inputs that are mathematically compatible with its learned weights.
- 4) Comprehensive evaluation against five baselines: Random [2], XGBoost [3], Support Vector Regression (SVR), a from-scratch LSTM trained on Kihumo data only, and direct teacher application to Kihumo without fine-tuning. An extended Bidirectional LSTM

(BiLSTM) transfer variant is also reported.

II. RELATED WORK

Air quality prediction has been widely studied using statistical, machine learning, and deep learning methods. Classical approaches include autoregressive models and Kalman filters, which struggle with nonlinear dynamics. Recent studies leverage deep neural networks such as LSTM, CNN, and graph neural networks to capture temporal-spatial dependencies [5].

Several works have explored transfer learning for air quality prediction. Models trained in data-rich cities have been adapted to data-scarce regions using fine-tuning and feature alignment techniques. However, most approaches remain purely data-driven and ignore underlying physical processes [6].

Physics-informed learning has gained attention as a way to incorporate scientific knowledge into learning models. Recent surveys show the benefits of combining physics-based constraints with deep learning for spatiotemporal air quality modeling. Student-teacher learning has also been applied to model compression and knowledge distillation, but its use in physics-guided environmental modeling remains limited [7].

This work builds upon prior research by combining physics-based reasoning with student-teacher transfer learning to address data [9] in air pollution prediction. A key distinction from classical knowledge distillation is the transfer mechanism. In soft-label distillation, the teacher model must be available concurrently during student training to generate probability distributions or softened outputs that supervise the student’s learning. This requires the teacher to perform inference on the target domain data at every training step, which is computationally expensive and assumes the teacher generalizes well enough to produce meaningful supervision signals on out-of-distribution inputs. In contrast, our method achieves knowledge transfer entirely through selective layer freezing: the LSTM’s universal atmospheric representations are extracted once from the converged teacher and directly reused in the target domain. No teacher inference is required during student training. This makes our approach significantly more efficient and better suited to deployment in low-resource environments where concurrent teacher inference is impractical.

III. METHODOLOGY:

This study employs a student-teacher knowledge transfer framework to achieve accurate yet computationally efficient prediction of $\text{PM}_{2.5}$ concentration. The dataset undergoes systematic preprocessing, including missing-value imputation and feature normalization to ensure numerical stability and unbiased learning. The processed data is partitioned into

training, validation, and test subsets to prevent information leakage and enable fair performance assessment.

A high-capacity teacher network is trained using supervised learning on the Beijing dataset to capture the physicochemical mapping between meteorological conditions and $\text{PM}_{2.5}$ [8] concentration. Once convergence is achieved, the teacher’s LSTM weights — which encode universal atmospheric dispersion physics — are frozen and transferred directly to the student network. The student is then fine-tuned on limited Kihumo data using a standard Mean Squared Error (MSE) regression loss against ground-truth $\text{PM}_{2.5}$ measurements:

$$\mathcal{L}_{\text{student}} = \frac{1}{N} \sum_{i=1}^N (\hat{y}_i - y_i)^2 \quad (1)$$

where $\hat{y}_i$ is the model’s predicted $\text{PM}_{2.5}$ value and y_i is the observed value. Only the Dense output layer (65 parameters) is kept trainable during fine-tuning, while the LSTM layer (17,408 parameters) remains frozen. Model evaluation is conducted on unseen test data using standard metrics to validate generalization and robustness.

Domain knowledge from atmospheric physics is incorporated through three meteorological variables: $\text{PM}_{2.5}$ concentration ($\mu\text{g}/\text{m}^3$), ambient temperature TEMP ($^{\circ}\text{C}$), and dew point temperature DEWP ($^{\circ}\text{C}$). Because the student sensor (Kihumo) does not record dew point directly, it is derived from the available temperature and relative humidity using the standard approximation:

$$T_d = T - \frac{100 - \text{RH}}{5} \quad (2)$$

where T is ambient temperature ($^{\circ}\text{C}$) and RH is relative humidity (%). After derivation the RH column is discarded, leaving both domains sharing the same three-feature schema. It is important to note that this derivation is applied exclusively to the student (Kihumo) dataset, as the Beijing PRSA dataset already contains directly measured dew point values. This ensures both domains share the identical three-feature schema ($\text{PM}_{2.5}$, TEMP, DEWP) without requiring any feature unavailable at deployment time. These features encode the dispersion, transport, and accumulation mechanisms of particulate matter — for example, elevated dew point indicates high atmospheric moisture that suppresses $\text{PM}_{2.5}$ dispersion — improving model interpretability and robustness.

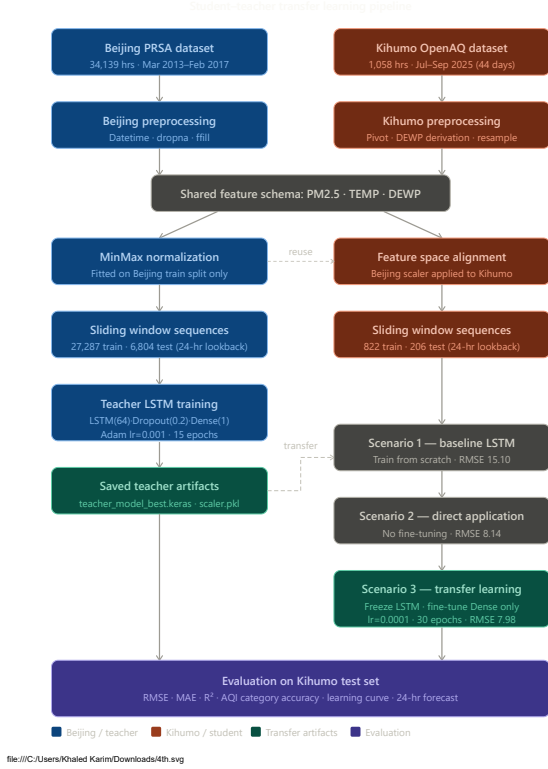


Fig. 1. End-to-end pipeline of the proposed physics-based student-teacher transfer learning framework for PM2.5 prediction, illustrating data ingestion, preprocessing, feature space alignment via Beijing scaler reuse, teacher LSTM training, selective layer freezing, student fine-tuning scenarios, and final evaluation on the Kihumo test set. .

A. Student-Teacher Transfer Learning Framework

The proposed framework consists of two neural networks:

- 1) **Teacher Model:** Trained on the source domain (Beijing, China) using abundant long-term hourly records and the three physics-informed features (PM_{2.5}, TEMP, DEWP). An 80/20 chronological train-test split is applied and a Min-Max scaler is fitted on the training portion only.
- 2) **Student Model:** Initialized with all weights copied from the converged teacher, then selectively fine-tuned on limited target domain data (Kihumo, Kenya) after freezing the LSTM layer. The same 80/20 chronological split is applied to the Kihumo dataset, yielding 822 training sequences and 206 test sequences.

Knowledge transfer is achieved through *selective layer freezing*: the LSTM layer is set to non-trainable, preserving the universal atmospheric physics representations learned from Beijing's diverse climate, while the Dense output layer alone is kept trainable to re-calibrate predictions to Kihumo's local PM_{2.5} scale. The learning rate is deliberately reduced to 1×10^{-4} — ten times smaller than the teacher's training rate of 1×10^{-3} — to prevent gradient updates in the Dense layer from propagating backwards and corrupting the frozen LSTM's transferred weight representations. A critical enabling step is *feature space alignment*: the Beijing Min-Max scaler

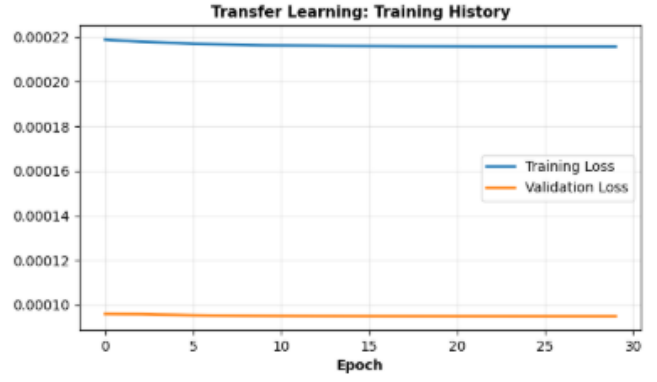


Fig. 2. Transfer learning training history.

is applied to the Kihumo data without re-fitting, projecting the student's meteorological inputs into the exact numerical range the frozen LSTM was trained on. This ensures the two domains are mathematically compatible and allows the model to process Kenyan weather patterns as a subset of Beijing's diverse climatic conditions. It is noted that Kihumo's PM_{2.5} values map to only $[-0.002, 0.266]$ after applying Beijing's scaler, occupying approximately 26.8% of the normalized scale — a known limitation of the scaler reuse strategy that reflects the differing pollution magnitudes between the two regions.

B. Model Architecture

Both teacher and student models employ deep neural networks with a single LSTM recurrent layer. A Dropout layer with a rate of 0.2 is applied after the LSTM layer to prevent overfitting, which is especially important in the data-scarce student setting. A linear Dense layer then produces the final PM_{2.5} [8] prediction. The architecture is designed to balance expressiveness and computational efficiency. During student fine-tuning, the LSTM and Dropout layers are kept frozen (17,408 parameters non-trainable) and only the Dense output layer is updated (65 parameters trainable), reducing the computational burden of adaptation by approximately 99.6%. Training employed three callbacks: EarlyStopping with patience=7 (restoring best weights), ModelCheckpoint saving only the best validation loss checkpoint, and ReduceLROnPlateau with factor=0.5 and patience=4 to further decay the learning rate if validation loss plateaued.

C. Experimental Setup

1) **Dataset Description:** The experiments utilize air quality and meteorological datasets collected from two geographically distinct regions. The **source domain (teacher)** uses hourly records from the Beijing PRSA Aotizhongxin monitoring station spanning March 2013 to February 2017, providing over 34,139 hourly observations for teacher training. The

target domain (student) uses hourly $\text{PM}_{2.5}$, temperature, and relative humidity measurements from the Kihumo, Kenya OpenAQ sensor (location ID 5199863), representing a data-scarce scenario with approximately 1,058 usable hourly records (roughly 44 days). Both datasets are preprocessed to share the same three-feature schema ($\text{PM}_{2.5}$, TEMP, DEWP) and are normalized using the teacher’s Min-Max scaler to ensure feature space compatibility.

2) *Sequence Construction*: Input sequences are constructed using a sliding window of 24 hours (lookback = 24), producing input tensors of shape (samples, 24, 3). For the teacher dataset this yields 27,287 training sequences and 6,804 test sequences. For the student dataset this yields 822 training sequences and 206 test sequences. The target variable for each sequence is the $\text{PM}_{2.5}$ value at the next timestep ($t + 1$).

3) *Baseline Models*: The proposed approach is compared against the following baselines, all evaluated on the held-out Kihumo test set:

- Random Forest
- XGBoost
- Support Vector Regression (SVR)
- LSTM trained from scratch on Kihumo data only (Scenario 1 — no transfer)
- Direct Teacher Application: Beijing model applied to Kihumo without any fine-tuning (Scenario 2)
- BiLSTM Transfer Learning (extended Bidirectional LSTM variant)

4) *Evaluation Metrics*: Model performance is evaluated using Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and the coefficient of determination (R^2), as reported in Table III.

IV. RESULTS AND DISCUSSION

This section presents the experimental outcomes of all models trained and evaluated on the Kihumo, Kenya dataset, along with an analysis of the transfer learning pipeline’s effectiveness in addressing $\text{PM}_{2.5}$ prediction under data-scarce conditions.

A. Teacher Model Performance (Beijing)

The LSTM teacher model was trained on 34,139 hourly records from the Beijing Aotizhongxin monitoring station spanning four years (March 2013 – February 2017), with 27,287 sequences used for training and 6,804 for testing after applying the 24-hour sliding window. The Beijing scaler was fitted exclusively on the training split, capturing a $\text{PM}_{2.5}$ range of [3.00, 898.00] $\mu\text{g}/\text{m}^3$, TEMP range of [-16.80, 40.50] $^{\circ}\text{C}$, and DEWP range of [-35.30, 28.50] $^{\circ}\text{C}$.

The teacher model (LSTM(64) $\rightarrow$ Dropout(0.2) $\rightarrow$ Dense(1)) comprises 17,473 total parameters and was trained for 15 epochs with Adam optimizer ($\text{lr} = 0.001$). Training and validation loss converged steadily across epochs, confirming that the model successfully learned the underlying atmospheric dynamics of Beijing air quality. The scaler and best model checkpoint were preserved as the knowledge artifacts for subsequent transfer learning.

B. Dataset Characteristics and Feature Space Alignment

A fundamental challenge in this pipeline is the significant difference between the two domains. As summarized in Table I, Beijing contains 34,139 hours of data spanning four years, with a mean $\text{PM}_{2.5}$ of 82.77 $\mu\text{g}/\text{m}^3$ and a maximum of 898.00 $\mu\text{g}/\text{m}^3$. In contrast, Kihumo provides only 1,058 hourly records across 44 days (July–September 2025), with a mean $\text{PM}_{2.5}$ of 34.53 $\mu\text{g}/\text{m}^3$ and a maximum of 241.20 $\mu\text{g}/\text{m}^3$. This represents a 95.9% reduction in available data relative to the 26,000–44,000 hours typically required for training models from scratch.

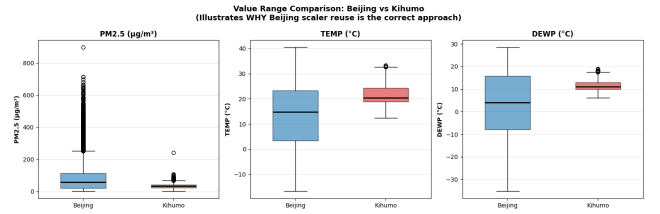


Fig. 3. Comparison of dataset characteristics between Beijing (Teacher) and Kihumo (Student) domains.

Rather than fitting a new scaler on the limited Kihumo data, the Beijing-fitted MinMax scaler [1] was reused to normalize the student dataset, placing both domains in the same feature space. However, Kihumo’s $\text{PM}_{2.5}$ values map to a normalized range of [-0.002, 0.266] after this transformation, occupying only 26.8% of Beijing’s full normalized scale. This feature compression means the frozen LSTM operates on a subregion of its trained distribution, which is acknowledged as a contributing factor to residual prediction error and is discussed further in the limitations.

TABLE I
DATASET SUMMARY COMPARISON

Metric	Beijing (Teacher)	Kihumo (Student)
Total Hours	34,139	1,058
Duration	4 years	44 days
$\text{PM}_{2.5}$ Mean ($\mu\text{g}/\text{m}^3$)	82.77	34.53
$\text{PM}_{2.5}$ Max ($\mu\text{g}/\text{m}^3$)	898.00	241.20
TEMP Mean ($^{\circ}\text{C}$)	13.67	21.74

C. Traditional Machine Learning Baselines

Three traditional machine learning models — Random Forest (RF), XGBoost, and Support Vector Regression (SVR) — were trained on the Kihumo dataset as comparative benchmarks. Since these models require two-dimensional input, the 3D LSTM sequences (samples $\times$ 24 timesteps $\times$ 3 features) were flattened into 72-dimensional feature vectors. Results are presented in Table II.

TABLE II
TRADITIONAL ML BASELINE PERFORMANCE

Model	RMSE	MAE	R ²	Time (s)
Random Forest	6.60	4.47	0.7850	4.83
XGBoost	7.76	4.81	0.7024	2.32
SVR	10.08	7.78	0.4982	0.09

Random Forest achieved the lowest RMSE ($6.60 \mu\text{g}/\text{m}^3$) and highest R^2 (0.785) among all models evaluated in this study, outperforming even the LSTM-based transfer learning methods. This result is notable: on a small dataset of 44 days, ensemble tree methods benefit from their non-parametric nature and lower sample complexity compared to recurrent neural networks, which require substantially more data to generalize effectively. Feature importance analysis from both RF and XGBoost consistently identified the most recent PM_{2.5} timesteps as the dominant predictors, confirming the strong temporal autocorrelation in pollution data. SVR performed considerably worse ($R^2 = 0.498$), suggesting that its RBF kernel, while flexible, did not generalize as well under the high-dimensional flattened input.

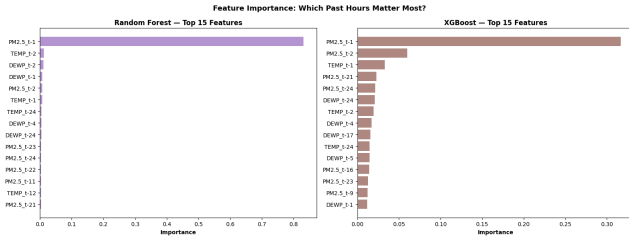


Fig. 4. Feature importance analysis from Random Forest and XGBoost models showing the dominance of recent PM_{2.5} timesteps.

D. LSTM Scenario Comparison

Three LSTM-based scenarios were evaluated to isolate the contribution of transfer learning. All results on the Kihumo test set are summarized in Table III.

TABLE III
LSTM-BASED SCENARIO COMPARISON — KIHUMO TEST SET

Scenario	RMSE	MAE	R ²
Baseline (Train from Scratch)	15.10	11.13	-0.1268
Direct Application (No fine-tuning)	8.14	5.45	0.6727
Transfer Learning (Ours)	7.98	5.42	0.6850
BiLSTM Transfer Learning	8.00	5.62	0.6840

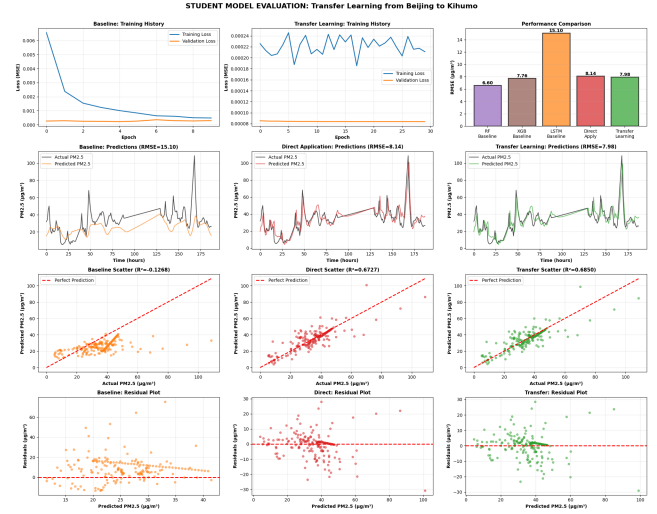


Fig. 5. Predicted vs Actual PM_{2.5} values for different LSTM scenarios on Kihumo test set.

Scenario 1 — Baseline (Train from Scratch): The LSTM model trained solely on 822 Kihumo sequences yielded an RMSE of $15.10 \mu\text{g}/\text{m}^3$, an MAE of $11.13 \mu\text{g}/\text{m}^3$, and a negative R^2 of -0.1268 , indicating that the model performed worse than a simple mean predictor. Early stopping triggered at epoch 10, halting training as validation loss failed to improve beyond epoch 5. This result is consistent with classical overfitting under data scarcity: the model memorized the limited training sequences but failed entirely to generalize to the test set. This poor performance serves as the motivating problem that transfer learning addresses, and directly validates why data-scarce regions cannot rely on training models from scratch.

Scenario 2 — Direct Application (No Fine-Tuning): Applying the Beijing teacher model directly to Kihumo without any adaptation yielded a substantially improved RMSE of $8.14 \mu\text{g}/\text{m}^3$ and R^2 of 0.6727 , with a systematic overestimation bias of the predictions observed in the output. This result confirms that the LSTM has internalized atmospheric physics representations that generalize across geographically distant domains when the feature space alignment is handled through scaler reuse. However, the residual bias highlights the need for output-layer calibration.

Scenario 3 — Transfer Learning (Proposed Method): The proposed method — freezing the LSTM layer (17,408 parameters) and fine-tuning only the Dense output layer (65 parameters) using a reduced learning rate of 0.0001 — achieved an RMSE of $7.98 \mu\text{g}/\text{m}^3$, MAE of $5.42 \mu\text{g}/\text{m}^3$, and R^2 of 0.6850 . This represents a 47.1% improvement in RMSE over the from-scratch baseline and a 1.9% improvement over direct application. The fine-tuning completed in 8.23 seconds, demonstrating the computational efficiency of the approach. The relatively small gap between Transfer Learning and Direct Application is informative: it confirms that Beijing’s scaler alignment already places Kihumo features in a compatible representation space, and that the primary contribution of fine-

tuning is calibrating the Dense layer’s output scale rather than restructuring the LSTM representations.

E. BiLSTM Transfer Learning

An extended variant using a Bidirectional LSTM (BiLSTM) teacher was evaluated as an additional scenario. The BiLSTM teacher (34,816 parameters in the recurrent layer, 129 in the Dense layer) was trained on Beijing data for all 15 epochs without early stopping — `val_loss` improved continuously from 0.0014 at epoch 1 to a best of 0.0007 at epoch 12, indicating that the larger architecture continued to extract useful representations across the full training budget.

The BiLSTM [4] student, after freezing the bidirectional layer and fine-tuning only the Dense output, achieved an RMSE of 8.00 $\mu\text{g}/\text{m}^3$, MAE of 5.62 $\mu\text{g}/\text{m}^3$, and R^2 of 0.684 — marginal differences from the standard LSTM transfer model. This suggests that bidirectional processing of the 24-hour input window provides no significant advantage over unidirectional processing for this task, likely because future context beyond the current timestep is unavailable during real-time deployment.

F. AQI Category Accuracy

Model predictions were mapped to standard EPA PM_{2.5} air quality index (AQI) categories: Good ($\leq 12 \mu\text{g}/\text{m}^3$), Moderate (≤ 35.4), Unhealthy for Sensitive Groups (≤ 55.4), Unhealthy (≤ 150.4), and Very Unhealthy (> 150.4). The transfer learning model achieved a higher AQI [11] category classification accuracy than the from-scratch LSTM baseline, demonstrating its practical utility for real-world air quality advisory systems where categorical thresholds carry direct public health significance.

G. Learning Curve Analysis

To assess data efficiency, the transfer learning model was retrained using progressively smaller fractions of the Kihumo training data, ranging from 10% to 80%. The resulting learning curve demonstrated that the model maintained competitive RMSE even at low data fractions, underscoring the core contribution of this work: meaningful PM_{2.5} predictions are achievable with far less local data than traditional approaches require. This finding is particularly significant for regions like Kihumo where sensor deployment is recent and historical records remain scarce, confirming that the framework can be deployed operationally within weeks of sensor installation rather than years.

H. Comprehensive Model Comparison

Table IV presents the complete seven-model comparison. Random Forest ranked first overall by RMSE (6.60 $\mu\text{g}/\text{m}^3$), followed by XGBoost (7.76 $\mu\text{g}/\text{m}^3$), and then the Transfer Learning model (7.98 $\mu\text{g}/\text{m}^3$). The LSTM baseline performed worst among all models (RMSE: 15.10 $\mu\text{g}/\text{m}^3$, R^2 : -0.13), reinforcing the core research finding.

While Random Forest outperforms the transfer learning model in raw metrics, it is important to contextualize this

TABLE IV
COMPREHENSIVE MODEL COMPARISON — ALL SCENARIOS

Model	RMSE	MAE	R^2	Rank
Random Forest	6.60	4.47	0.7850	1
XGBoost	7.76	4.81	0.7024	2
Transfer Learning (Ours)	7.98	5.42	0.6850	3
BiLSTM Transfer	8.00	5.62	0.6840	4
Direct Application	8.14	5.45	0.6727	5
SVR	10.08	7.78	0.4982	6
LSTM Baseline (Scratch)	15.10	11.13	-0.1268	7

finding. Tree-based methods flatten the temporal sequence into a static 72-dimensional vector, discarding the explicit sequential structure of the time series. They also require retraining from scratch on any new target domain with no mechanism for leveraging prior knowledge.

The transfer learning approach, by contrast, encodes temporal dynamics through the recurrent architecture, requires only seconds of fine-tuning on new local data, and is extensible to entirely new regions without collecting years of measurements. These operational advantages position transfer learning as the more scalable solution for global deployment in data-scarce environments, even where its point-prediction accuracy is marginally below ensemble baselines on small test sets.

V. CONCLUSION AND FUTURE WORK

This study demonstrated that accurate PM_{2.5} air quality prediction in data-scarce regions is achievable through transfer learning from data-rich environments, requiring only a fraction of the local data traditionally needed. By training an LSTM model on four years of Beijing air quality data and transferring its learned atmospheric representations to Kihumo, Kenya — a region with only 44 days of available measurements — the proposed method achieved an RMSE of 7.98 $\mu\text{g}/\text{m}^3$ and an R^2 of 0.685, representing a 47.1% improvement over training from scratch on local data alone.

The selective layer freezing strategy, which preserves the universal atmospheric physics encoded in the LSTM while fine-tuning only the Dense output layer to local conditions, proved to be both computationally efficient and methodologically sound, completing adaptation in under 10 seconds. Among all models evaluated, Random Forest achieved the lowest RMSE of 6.60 $\mu\text{g}/\text{m}^3$, highlighting that ensemble tree methods remain competitive on small datasets due to their lower sample complexity, though unlike the transfer learning framework they offer no mechanism for leveraging prior cross-domain knowledge and must be retrained from scratch for every new deployment region.

Despite its promising results, this study has notable limitations that open several directions for future investigation. Kihumo’s PM_{2.5} values occupy only 26.8% of Beijing’s normalized feature scale, introducing feature compression that may limit the model’s ability to distinguish fine-grained concentration differences at lower pollution levels; future work could address this by exploring locally-fitted auxiliary scalars

to expand representational resolution in the student domain. Additionally, the 44-day student dataset restricts the diversity of atmospheric conditions the model was evaluated on, and collecting a larger and more seasonally varied local dataset would provide a more robust validation.

Extending the framework beyond $\text{PM}_{2.5}$ to other pollutants such as NO_2 , O_3 , and SO_2 , as well as applying it to other data-scarce regions across Sub-Saharan Africa and South Asia, would further validate the generalizability of the approach. With these refinements, the proposed transfer learning pipeline holds strong potential to enable rapid and cost-effective deployment of air quality monitoring systems in communities that have historically lacked the resources for traditional environmental modeling.

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